

Tide retrospective: the numerator expansion lands outer tails and the packaging of stage 5

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Abstract

The numerator stage of the forward-expansion programme is complete: for every continuous observable P of polynomial growth,

$$\int P(z) e^{-(L(qz)-L(0))/q^2} \mathbf{1}_{\{qz \in U\}} dz = \sum_{j=0}^N a_j q^j + o(q^N) \quad (q \rightarrow 0^+),$$

with $a_j = \int P e^{-T_2} P_j$ (`numerator_hasExpansion`). This tide supplies what the window convergence of the previous tide left open: the two outer tails and the decomposition. The true integrand's tail needs no binomial gymnastics — the growth certificate $|P| \leq C_P(1 + \|z\|^n)$ splits into exactly two power-weighted tails of the stage-2 coercivity transfer. The coefficient polynomial's tail collapses to a single polynomial-Gaussian bound and splits binomially into the stage-2 Gaussian tails. The decomposition is four subidentities — numerator split, U -indicator collapse on the window (where q -scaled points land inside the domain ball), coefficient-sum/integral exchange, and the window difference — closed by linear arithmetic, with the window piece converted to $o(q^N)$ through `isLittle0_iff_tendsto` and the dominated-convergence theorem of the previous tide.

1 Setting

Stage 5c-ii-b, the consult's items 8–9, completing the numerator stage begun seven tides ago. Everything here is bookkeeping around already-proven analysis; the only judgement calls were where to put absolute values (inside the tail integrals, so the assembly needs just $|\int f| \leq \int |f|$) and to keep the two-term growth form $C_P(\|z\|^0 + \|z\|^n)$ for the true tail rather than reflexively binomially expanding.

2 The thing formalised

`abs_integrand_eq` (the integrand's absolute value factors through the observable), `observable_integrand_tail_isLittle0` and `coeff_polynomial_tail_isLittle0` (both beyond all orders), `integrable_coeff_polynomial`, and `numerator_hasExpansion` as a `Laplace.IsAsymptoticExpansionTo` with coefficients `numeratorCoeff`. The tails follow one template: an `isBig0` comparison (constant one) against a bound function assembled from stage-2 tail lemmas, using `setIntegral_mono_on` with explicit integrabilities on the complement, then `isBig0.trans_isLittle0`.

3 Where progress stalled

One recurring class, three sites: `integral_congr_ae` hands its pointwise goals as applied lambdas, and every `rw` inside dies on the beta redex — `beta_reduce` first, always. The style linter rejected two goal-changing `shows` (use `change`). A cross-worktree slip: a build command ran in the SRI root because the previous shell had died; the worktree path must be re-entered per command batch.

4 Mathlib lookup versus new mathematics

Lookup: `MeasureTheory.integral.add_compl`, `setIntegral_congr_fun`,
`setIntegral_mono_on`, `IntegrableOn.congr_fun`, `integral.indicator`,
`integral_finset_sum`, `norm_integral_le_integral_norm`, `isLittleO_iff_tendsto`.
 New mathematics: none; the stage’s design concentrated all analysis in the majorant and the DCT, and this tide cashed the cheque.

5 What the next tide inherits

Stage 6: the partition function is the $P = 1$ instance of `numerator_hasExpansion`; its constant coefficient $a_0 = \int e^{-T_2} > 0$ (the coefficient functions at $j = 0$ are constantly one, and the Gaussian integral is positive); a generic finite asymptotic-division lemma (two order- N expansions, the denominator’s constant coefficient nonzero, yields the expansion of the quotient) then gives the *moment* expansion. Stage 7 states the public theorems: existence of the moment expansion for the monomial observables and the jet-comparison corollary through stage A1’s coefficient uniqueness — the germbij forward direction.